

AGRIVO - Complete Technical Documentation

Global Agricultural Services Marketplace

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1. GENERAL DESCRIPTION

Agrivo is a global marketplace connecting farmers with field professionals. The platform enables:

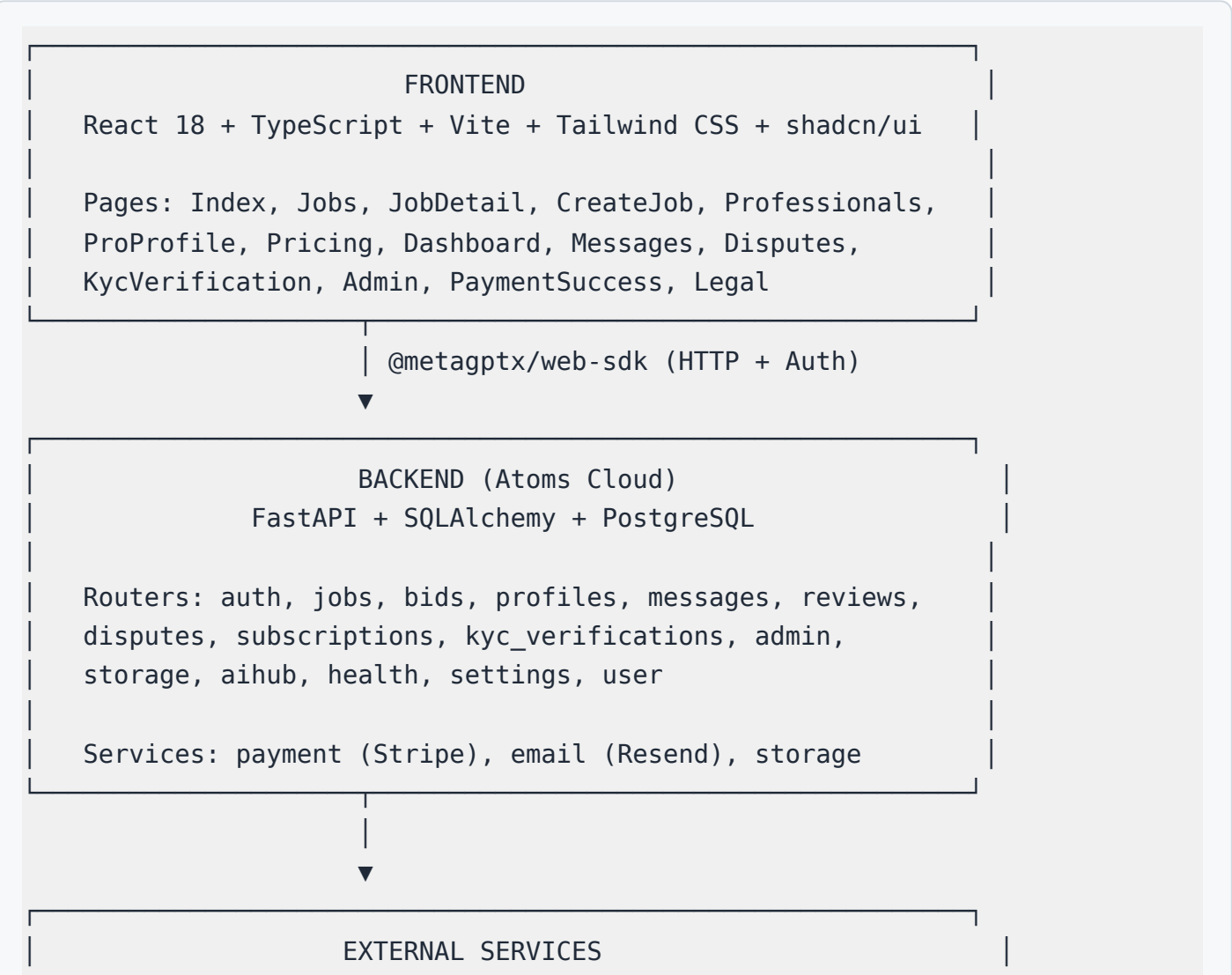
- **Post agricultural jobs** with reverse auction or fixed price system
- **Professional directory** with KYC verification
- **Bidding system** where professionals compete for jobs
- **Real-time messaging** between users
- **Reviews and ratings** for professionals
- **Subscription plans** (Free, Pro €19/month, Enterprise €29/month) with Stripe
- **KYC verification** post-payment with admin panel
- **Dispute resolution center** for conflict management
- **Multi-language support** (Spanish, English, Portuguese, French)

- **10 operating countries:** Spain, Brazil, Argentina, USA, Portugal, France, India, Australia, Ukraine, Mexico
- **12 service categories:** Drones, Harvesting, Pruning, Plowing, Seeding, Irrigation, Spraying, Soil Analysis, Topography, Livestock, Consulting, Transport

Business Model

Plan	Price	Features
Free	€0	3 jobs/month, 5 bids/job, basic profile
Professional	€19/month	Verified KYC, ✓ badge, unlimited bids, priority ranking
Enterprise	€29/month	Featured on homepage, Top Pro ★ badge, analytics, priority support

2. SYSTEM ARCHITECTURE



- Stripe (Payments & subscriptions)
- Resend (Email notifications)
- Object Storage (KYC photos, avatars)

Main Data Flow

1. **User posts job** → Frontend → SDK → Backend → PostgreSQL
2. **Professional bids** → Frontend → SDK → Backend → PostgreSQL + Email notification
3. **Subscription payment** → Frontend → Stripe Checkout → Webhook → Backend → Updates plan
4. **KYC** → Frontend (form + photo) → Backend → Admin approves/rejects → Email

3. TECHNOLOGY STACK

Frontend

Technology	Version	Usage
React	18.3.1	UI Framework
TypeScript	5.5.3	Static typing
Vite	5.4.1	Build tool
Tailwind CSS	3.4.11	Utility-first styles
shadcn/ui	latest	UI Components (Radix)
React Router	6.30.0	SPA routing
TanStack Query	5.56.2	Cache & fetching
Recharts	2.12.7	Charts
Lucide React	0.462.0	Icons
@metagptx/web-sdk	latest	Backend SDK
Sonner	1.7.4	Toast notifications
React Hook Form + Zod	latest	Forms & validation

Backend

Technology	Usage
FastAPI	REST API framework
SQLAlchemy	PostgreSQL ORM
Alembic	DB migrations
PostgreSQL	Relational database
Stripe SDK	Payment processing
Resend	Email delivery

Visual Design

Aspect	Value
Primary color	Emerald/Green (#059669, #166534)
Secondary color	Earth/Amber (#92400e)
Accent color	Orange (#ea580c)
Background	Warm white (#fafaf5)
Heading font	Poppins
Body font	Open Sans
Style	Trust & Authority, marketplace

4. DATABASE - SCHEMAS

Table: `jobs` (Jobs/Work Orders)

```
{
  "title": "jobs",
  "type": "object",
  "properties": {
```

```

    "id": { "type": "integer", "autoincrement": true },
    "title": { "type": "string" },
    "description": { "type": "string" },
    "category": { "type": "string" },
    "country": { "type": "string" },
    "location": { "type": "string" },
    "hectares": { "type": "number" },
    "budget_min": { "type": "number" },
    "budget_max": { "type": "number" },
    "contract_type": { "type": "string", "enum": ["reverse_auction",
"fixed_price"] },
    "status": { "type": "string", "default": "open", "enum": ["open",
"in_progress", "completed", "cancelled"] },
    "user_id": { "type": "string" }
  },
  "required": ["title", "category", "country", "location", "contract_type",
"user_id"],
  "auto_fields": ["created_at", "updated_at"]
}

```

Table: **bids** (Bids/Offers)

```

{
  "title": "bids",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },
    "job_id": { "type": "integer" },
    "amount": { "type": "number" },
    "message": { "type": "string" },
    "status": { "type": "string", "default": "pending", "enum": ["pending",
"accepted", "rejected"] },
    "user_id": { "type": "string" }
  },
  "required": ["job_id", "amount", "user_id"]
}

```

Table: **profiles** (User Profiles)

```

{
  "title": "profiles",
  "type": "object",

```

```

"properties": {
  "id": { "type": "integer" },
  "display_name": { "type": "string" },
  "role": { "type": "string", "enum": ["farmer", "professional"] },
  "country": { "type": "string" },
  "description": { "type": "string" },
  "avatar_url": { "type": "string" },
  "rating": { "type": "number", "default": 0 },
  "jobs_completed": { "type": "integer", "default": 0 },
  "service_radius_km": { "type": "integer", "default": 50 },
  "verified_kyc": { "type": "boolean", "default": false },
  "categories": { "type": "string", "description": "comma-separated categories" },
  "language": { "type": "string", "default": "es" },
  "user_id": { "type": "string" }
},
"required": ["display_name", "role", "user_id"]
}

```

Table: messages (Messages)

```

{
  "title": "messages",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },
    "job_id": { "type": "integer" },
    "sender_id": { "type": "string" },
    "receiver_id": { "type": "string" },
    "content": { "type": "string" },
    "user_id": { "type": "string" }
  },
  "required": ["job_id", "content", "user_id"]
}

```

Table: reviews (Reviews)

```

{
  "title": "reviews",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },

```

```

    "professional_id": { "type": "string" },
    "job_id": { "type": "integer" },
    "rating": { "type": "integer", "min": 1, "max": 5 },
    "comment": { "type": "string" },
    "reviewer_name": { "type": "string" },
    "user_id": { "type": "string" }
  },
  "required": ["professional_id", "rating", "comment", "user_id"]
}

```

Table: disputes (Disputes)

```

{
  "title": "disputes",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },
    "job_title": { "type": "string" },
    "reason": { "type": "string" },
    "description": { "type": "string" },
    "amount_disputed": { "type": "number" },
    "status": { "type": "string", "enum": ["open", "in_review", "resolved",
"closed"] },
    "resolution": { "type": "string" },
    "user_id": { "type": "string" }
  },
  "required": ["job_title", "reason", "description", "status", "user_id"]
}

```

Table: subscriptions (Subscriptions)

```

{
  "title": "subscriptions",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },
    "plan": { "type": "string", "enum": ["free", "pro", "enterprise"] },
    "status": { "type": "string", "default": "active" },
    "stripe_session_id": { "type": "string" },
    "user_id": { "type": "string" }
  },
}

```

```
"required": ["plan", "user_id"]
}
```

Table: kyc_verifications (KYC Verifications)

```
{
  "title": "kyc_verifications",
  "type": "object",
  "properties": {
    "id": { "type": "integer" },
    "full_name": { "type": "string" },
    "document_type": { "type": "string", "enum": ["dni", "passport",
"license"] },
    "document_number": { "type": "string" },
    "country": { "type": "string" },
    "address": { "type": "string" },
    "specialty": { "type": "string" },
    "years_experience": { "type": "integer" },
    "certifications": { "type": "string" },
    "description": { "type": "string" },
    "document_photo_url": { "type": "string" },
    "status": { "type": "string", "enum": ["pending", "approved", "rejected"]
},
    "plan": { "type": "string" },
    "user_id": { "type": "string" }
  },
  "required": ["full_name", "document_type", "document_number", "status",
"user_id"]
}
```

5. BACKEND - API & SERVICES

Main Endpoints

Method	Route	Description
GET	/api/v1/auth/login	Initiate OAuth login
GET	/api/v1/auth/me	Get current user

GET	/api/v1/auth/logout	Logout
GET/POST	/api/v1/jobs/	List/Create jobs
GET	/api/v1/jobs/{id}	Job detail
GET/POST	/api/v1/bids/	List/Create bids
GET/POST	/api/v1/profiles/	List/Create profiles
GET/POST	/api/v1/messages/	List/Send messages
GET/POST	/api/v1/reviews/	List/Create reviews
GET/POST	/api/v1/disputes/	List/Create disputes
GET/POST	/api/v1/subscriptions/	List/Create subscriptions
GET/POST	/api/v1/kyc_verifications/	List/Create KYC verifications
POST	/api/v1/admin/kyc/{id}/approve	Approve KYC
POST	/api/v1/admin/kyc/{id}/reject	Reject KYC
POST	/api/v1/storage/upload	Upload files
GET	/api/v1/health	Health check

Email Service (Resend)

```
# app/backend/services/email_service.py
# Sends email notifications for:
# - KYC approved/rejected
# - New bid received
# - New message received
```

Payment Service (Stripe)

```
# app/backend/services/payment.py
# - create_payment_session(): Creates Stripe checkout session
# - verify_payment(): Verifies completed payment
# Plans: Pro (€19/month), Enterprise (€29/month)
```

Backend Router Structure

```
app/backend/
├─ main.py                # FastAPI entry point
├─ lambda_handler.py      # Serverless deploy handler
├─ core/
│   ├─ auth.py            # Auth configuration
│   ├─ config.py          # Environment variables
│   ├─ database.py        # PostgreSQL connection
│   └─ enums.py           # Enumerations
├─ models/                # SQLAlchemy models
│   ├─ jobs.py
│   ├─ bids.py
│   ├─ profiles.py
│   ├─ messages.py
│   ├─ reviews.py
│   ├─ disputes.py
│   ├─ subscriptions.py
│   └─ kyc_verifications.py
├─ routers/              # API endpoints
│   ├─ admin.py           # Admin KYC panel
│   ├─ auth.py            # Authentication
│   ├─ jobs.py
│   ├─ bids.py
│   ├─ profiles.py
│   ├─ messages.py
│   ├─ reviews.py
│   ├─ disputes.py
│   ├─ subscriptions.py
│   ├─ kyc_verifications.py
│   ├─ storage.py         # File upload
│   └─ aihub.py           # AI capabilities
├─ services/             # Business logic
│   ├─ email_service.py   # Emails with Resend
│   ├─ payment.py         # Payments with Stripe
│   ├─ storage.py         # Object storage
│   └─ mock_data.py       # Initial data
└─ data_models/          # Table JSON schemas
    ├─ jobs.json
    ├─ bids.json
    ├─ profiles.json
    ├─ messages.json
    ├─ reviews.json
    └─ disputes.json
```

- └─ subscriptions.json
- └─ kyc_verifications.json

6. FRONTEND - VIEWS & COMPONENTS

Application Routes

Route	Component	Description
/	Index.tsx	Landing page with hero, stats, categories, featured pros, jobs
/jobs	Jobs.tsx	Job listing with filters (category, country, type)
/jobs/new	CreateJob.tsx	Form to post a new job
/jobs/:id	JobDetail.tsx	Job detail with bidding system
/pros	Professionals.tsx	Professional directory with filters
/pros/:id	ProProfile.tsx	Professional public profile with reviews
/precios	Pricing.tsx	Subscription plans with Stripe checkout
/dashboard	Dashboard.tsx	User panel (my jobs, my bids, profile)
/messages	Messages.tsx	Messaging system between users
/disputes	Disputes.tsx	Dispute resolution center
/payment-success	PaymentSuccess.tsx	Successful payment confirmation
/kyc	KycVerification.tsx	KYC verification form
/admin	Admin.tsx	Admin panel (KYC management)
/auth/callback	AuthCallback.tsx	OAuth callback
/auth/error	AuthError.tsx	Authentication error
/legal/:page	Legal.tsx	Legal pages (privacy, terms, cookies)

Shared Components

Component	Description
Header.tsx	Main navigation, language selector, login/logout, user menu
Footer.tsx	Links, categories, legal, social media
Badges.tsx	Verified (✓) and Top Pro (★) badges
CookieConsent.tsx	Cookie consent banner
LoadingSpinner.tsx	Loading indicator
ProtectedAdminRoute.tsx	HOC for admin routes

Landing Page (Index.tsx) - Sections

1. **Hero:** Background image with agricultural drone, animated title, CTAs, country flags
2. **Animated statistics:** Active professionals, completed jobs, countries, satisfaction
3. **How it works:** 3 steps (post → receive offers → hire)
4. **Categories:** Grid of 12 categories with emoji icons
5. **Featured professionals:** Cards with avatar, rating, badges, country
6. **Recent jobs:** Cards with budget, location, category
7. **Final CTA:** Call to action to register
8. **Footer:** Links, legal, language

Filter System (Jobs.tsx & Professionals.tsx)

- Text search
 - Category filter (12 options)
 - Country filter (10 options)
 - Contract type filter (reverse auction / fixed price)
 - Sort by date, budget, rating
-

7. AUTHENTICATION & SECURITY

Authentication Flow

1. User clicks "Sign In"
2. Frontend calls `authApi.login()`
3. Backend redirects to OAuth provider (OIDC)
4. User authenticates
5. Callback to `/auth/callback`
6. Backend sets session cookie
7. Frontend gets user with `client.auth.me()`

Auth Code (Frontend)

```
// src/lib/auth.ts
import axios from 'axios';
import { getAPIBaseURL } from './config';

class RPApi {
  private client: AxiosInstance;

  constructor() {
    this.client = axios.create({
      withCredentials: true,
      headers: { 'Content-Type': 'application/json' },
    });
  }

  async getCurrentUser() {
    const response = await
this.client.get(`${this.getBaseURL()}/api/v1/auth/me`);
    return response.data;
  }

  async login() {
    const response = await
this.client.get(`${this.getBaseURL()}/api/v1/auth/login`);
    window.location.href = response.data.redirect_url;
  }

  async logout() {
    const response = await
```

```
this.client.get(`${this.getBaseURL()}/api/v1/auth/logout`);
  window.location.href = response.data.redirect_url;
}
}
```

SDK for Entities

```
// src/lib/api.ts
import { createClient } from '@metagptx/web-sdk';
export const client = createClient();

// Usage in components:
// client.entities.jobs.queryAll({ country: 'España' })
// client.entities.jobs.queryMine()
// client.entities.jobs.create({ title: '...', ... })
// client.entities.bids.create({ job_id: 1, amount: 5000 })
// client.payment.createPaymentSession({ ... })
// client.payment.verifyPayment(sessionId)
```

8. PAYMENT SYSTEM (STRIPE)

Payment Flow

1. User selects plan (Pro €19 or Enterprise €29)
2. Frontend calls `client.payment.createPaymentSession()`
3. Stripe generates checkout URL
4. User completes payment on Stripe
5. Redirect to `/payment-success?session_id=xxx`
6. Frontend verifies with `client.payment.verifyPayment()`
7. Backend updates user subscription
8. Redirect to `/kyc` for identity verification

Proration (Plan Upgrade)

- If user already has Pro plan and wants Enterprise, proration is applied
- Remaining credit from current plan is calculated
- Only the difference is charged

9. INTERNATIONALIZATION (i18n)

Supported Languages

Code	Language	Flag
es	Spanish	
en	English	
pt	Portuguese	
fr	French	

Implementation

```
// src/lib/i18n.ts
// localStorage-based system with automatic browser detection
// Usage: const label = t('nav.jobs'); // "Jobs" in English
// Hook: const [locale, setLocale] = useLocale();
```

Main Translation Keys

- `nav.*` - Navigation
- `hero.*` - Hero section
- `stats.*` - Statistics
- `how.*` - How it works
- `categories.*` - Categories
- `jobs.*` - Jobs page
- `pros.*` - Professionals page
- `pricing.*` - Pricing page
- `footer.*` - Footer

10. SEED DATA

Spanish Professionals (14 profiles)

Name	Specialty	Location	Rating	Plan
María García López	Pruning	La Rioja	4.8	Pro
Miguel Ángel Ruiz	Plowing	Andalusia	4.6	Enterprise
Carlos Martínez Herrera	Irrigation	Jaén, Andalusia	4.9	Enterprise
Laura Sánchez Moreno	Drones	Castilla-La Mancha	4.8	Pro
Javier Fernández Ortega	Topography	Aragon/Catalonia	4.7	Pro
Ana Rodríguez Vega	Soil Analysis	Valencia	4.9	Enterprise
Pedro Navarro Gil	Harvesting	Castilla y León	4.6	Pro
Isabel Torres Muñoz	Livestock	Extremadura	4.8	Enterprise
Francisco López Castillo	Spraying	Andalusia	4.5	Pro
Carmen Díaz Romero	Consulting	Andalusia	4.9	Enterprise
Antonio Gómez Serrano	Seeding	Castilla-La Mancha	4.7	Pro
Elena Martín Blanco	Pruning	Murcia/Almería	4.8	Pro
David Ruiz Peña	Transport	All Spain	4.6	Enterprise
Sofía Hernández Rivas	Drones	Lleida	4.8	Pro

Spanish Jobs (14 listings)

Job	Location	Hectares	Budget
Pruning 3,000 olive trees	Jaén, Andalusia	45	€8,000-12,000
Drip irrigation for olive grove	Jaén, Andalusia	45	€8,000-15,000
Drone spraying in vineyard	Haro, La Rioja	30	€3,000-5,500
Soil analysis for citrus	Alzira, Valencia	20	€2,000-4,000

Cereal harvesting	Medina del Campo, Valladolid	200	€12,000-20,000
Super-intensive olive pruning	Lucena, Córdoba	60	€5,000-9,000
Livestock consulting in dehesa	Trujillo, Cáceres	150	€3,000-6,000
Topography and leveling	Monzón, Huesca	80	€4,000-7,000
Direct sunflower seeding	Manzanares, Ciudad Real	100	€5,000-8,000
Almond tree fertilization	Cieza, Murcia	35	€2,500-4,500
Hydroponic greenhouse irrigation	El Ejido, Almería	5	€15,000-25,000
Ground cereal spraying	Toro, Zamora	150	€3,500-5,000
Olive transport	Baena, Córdoba	80	€4,000-6,000
Plowing for vineyard	Aranda de Duero, Burgos	25	€6,000-9,000

11. DEPLOYMENT

Required Environment Variables

```
# Frontend
VITE_API_BASE_URL=<backend URL>

# Backend
DATABASE_URL=postgresql://user:pass@host:5432/dbname
STRIPE_SECRET_KEY=sk_live_XXXXX
STRIPE_WEBHOOK_SECRET=whsec_XXXXX
RESEND_API_KEY=re_XXXXX
JWT_SECRET=<secret>
OIDC_CLIENT_ID=<client_id>
OIDC_CLIENT_SECRET=<client_secret>
OIDC_ISSUER_URL=<issuer_url>
```

Build Commands

```
# Frontend
cd app/frontend
```

```
pnpm install
pnpm run build    # Generates dist/

# Backend
cd app/backend
pip install -r requirements.txt
alembic upgrade head    # Migrations
uvicorn main:app --host 0.0.0.0 --port 8000
```

12. FILE STRUCTURE

```
app/
├── backend/
│   ├── main.py
│   ├── lambda_handler.py
│   ├── alembic/
│   │   └── versions/          # Migrations
│   ├── core/
│   │   ├── auth.py
│   │   ├── config.py
│   │   ├── database.py
│   │   └── enums.py
│   ├── data_models/          # JSON schemas
│   │   ├── jobs.json
│   │   ├── bids.json
│   │   ├── profiles.json
│   │   ├── messages.json
│   │   ├── reviews.json
│   │   ├── disputes.json
│   │   ├── subscriptions.json
│   │   └── kyc_verifications.json
│   ├── models/               # SQLAlchemy models
│   │   ├── jobs.py
│   │   ├── bids.py
│   │   ├── profiles.py
│   │   ├── messages.py
│   │   ├── reviews.py
│   │   ├── disputes.py
│   │   ├── subscriptions.py
│   │   └── kyc_verifications.py
│   └── routers/              # API endpoints
```

```

├── admin.py
├── auth.py
├── jobs.py
├── bids.py
├── profiles.py
├── messages.py
├── reviews.py
├── disputes.py
├── subscriptions.py
├── kyc_verifications.py
├── storage.py
├── aihub.py
├── services/ # Business logic
│   ├── email_service.py
│   ├── payment.py
│   ├── storage.py
│   └── mock_data.py
├── schemas/
│   ├── auth.py
│   ├── aihub.py
│   └── storage.py
├── frontend/
│   ├── index.html
│   ├── package.json
│   ├── vite.config.ts
│   ├── tailwind.config.ts
│   ├── tsconfig.json
│   ├── src/
│   │   ├── main.tsx
│   │   ├── App.tsx # Main router
│   │   ├── index.css # Global styles + theme
│   │   ├── components/
│   │   │   ├── Header.tsx
│   │   │   ├── Footer.tsx
│   │   │   ├── Badges.tsx
│   │   │   ├── CookieConsent.tsx
│   │   │   ├── LoadingSpinner.tsx
│   │   │   ├── ProtectedAdminRoute.tsx
│   │   │   └── ui/ # shadcn/ui components (40+)
│   │   ├── pages/
│   │   │   ├── Index.tsx # Landing page
│   │   │   ├── Jobs.tsx # Job listing
│   │   │   └── JobDetail.tsx # Detail + bids

```

```

├── CreateJob.tsx # Create job
├── Professionals.tsx # Directory
├── ProProfile.tsx # Professional profile
├── Pricing.tsx # Plans
├── Dashboard.tsx # User panel
├── Messages.tsx # Messaging
├── Disputes.tsx # Disputes
├── KycVerification.tsx # KYC form
├── Admin.tsx # Admin panel
├── PaymentSuccess.tsx
├── AuthCallback.tsx
├── AuthError.tsx
├── Legal.tsx
├── lib/
│   ├── api.ts # SDK client
│   ├── auth.ts # Auth API
│   ├── config.ts # Configuration
│   ├── constants.ts # Seed data, categories, countries
│   ├── i18n.ts # Internationalization
│   └── utils.ts # Utilities
├── contexts/
│   └── AuthContext.tsx
├── hooks/
│   ├── use-mobile.tsx
│   └── use-toast.ts
├── public/
└── assets/ # Static images

```

NOTES FOR THE DEVELOPER

1. **Web SDK:** The frontend uses `@metagptx/web-sdk` which abstracts backend calls. To replicate without this SDK, implement an HTTP client that handles auth with cookies and entity CRUD.
2. **Avatars:** Dynamically generated with DiceBear API:
<https://api.dicebear.com/7.x/initials/svg?seed={name}>
3. **Images:** Hero and category images are hosted on CDN. To replicate, use your own images or services like Unsplash.
4. **Stripe:** Requires a Stripe account with products configured for Pro and Enterprise plans.
5. **Resend:** For transactional emails. Alternative: SendGrid, Mailgun, or any SMTP service.

6. **Seed Data:** Initial data (professionals and jobs) is displayed as sample content until real users register. It merges with real DB data.

Document generated on July 11, 2026 Version: 1.0 Platform: Agrivo - Global Agricultural Services Marketplace